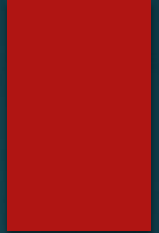


Stack

Stack



- ▶ Stack in real life: stack of books, stack of plates
- ▶ Add new item at the top
- ▶ Remove an item at the top
- ▶ **Can only access element at the top**

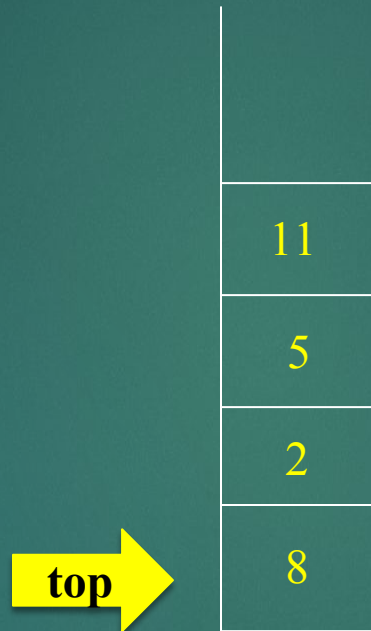


Interface methods of the Stacks



- ▶ **Push(X)** Insert at the top
- ▶ **Pop()** remove the top element
- ▶ **Top()** read the top element

push() Example:



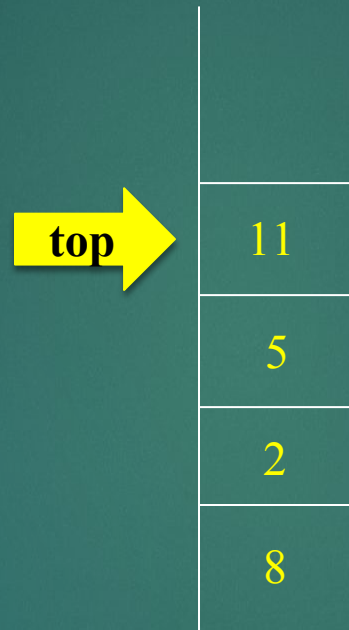
push(8)

push(2)

push(5)

push(11)

pop() Example:



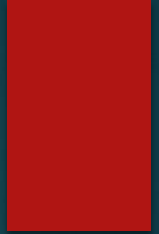
pop() $\square$ 8

pop() $\rightarrow$ 2

pop() $\square$ 5

pop() $\square$ 11

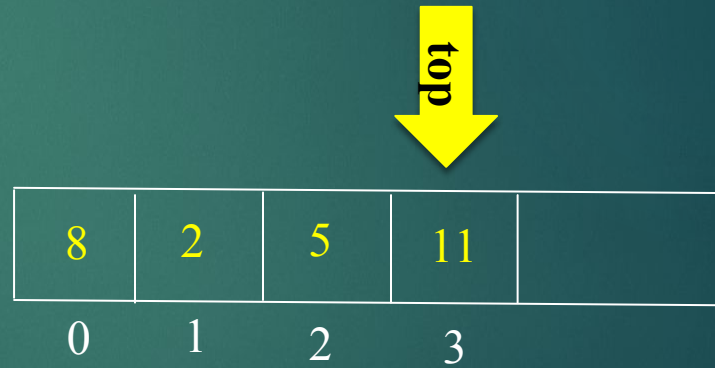
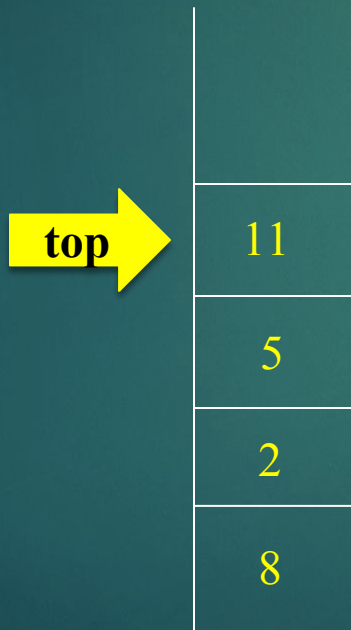
Stack(LIFO)



- ▶ LIFO(Last in First out)
- ▶ pop() on an empty Stack: solution isEmpty()
- ▶ isEmpty before pop developer responsibility

Stack Implementation using array

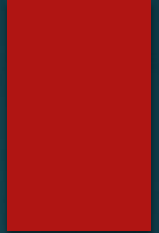
- ▶ In Array insert or delete at begin creates problem
 - ▶ Solution: Best way at the end of array



Stack Implementation using array

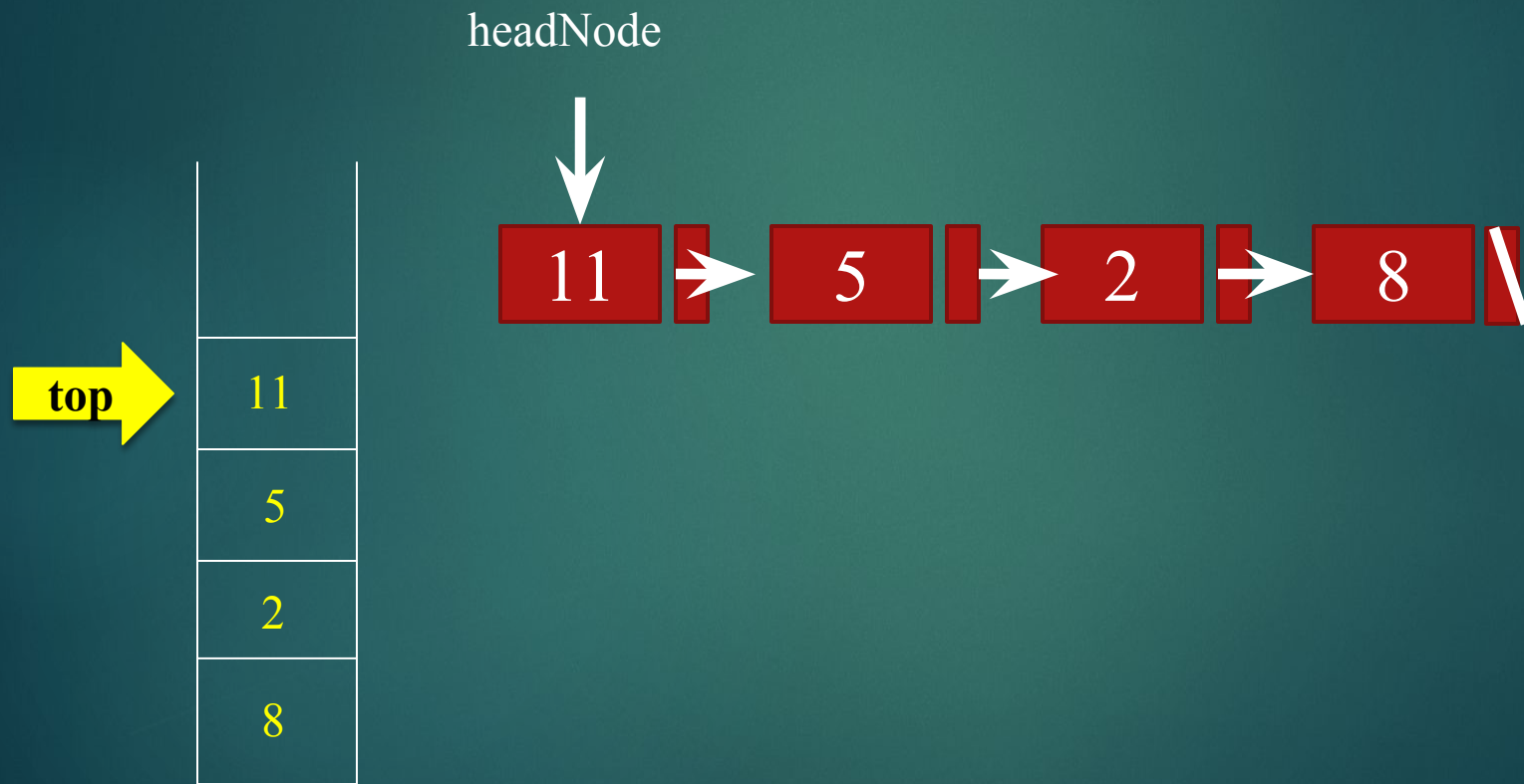
- ▶ In case of an array it is possible that the array may fill up if we push enough elements
- ▶ IsFull() return true stack(array) is full. isFull call before calling push

Stack using link list

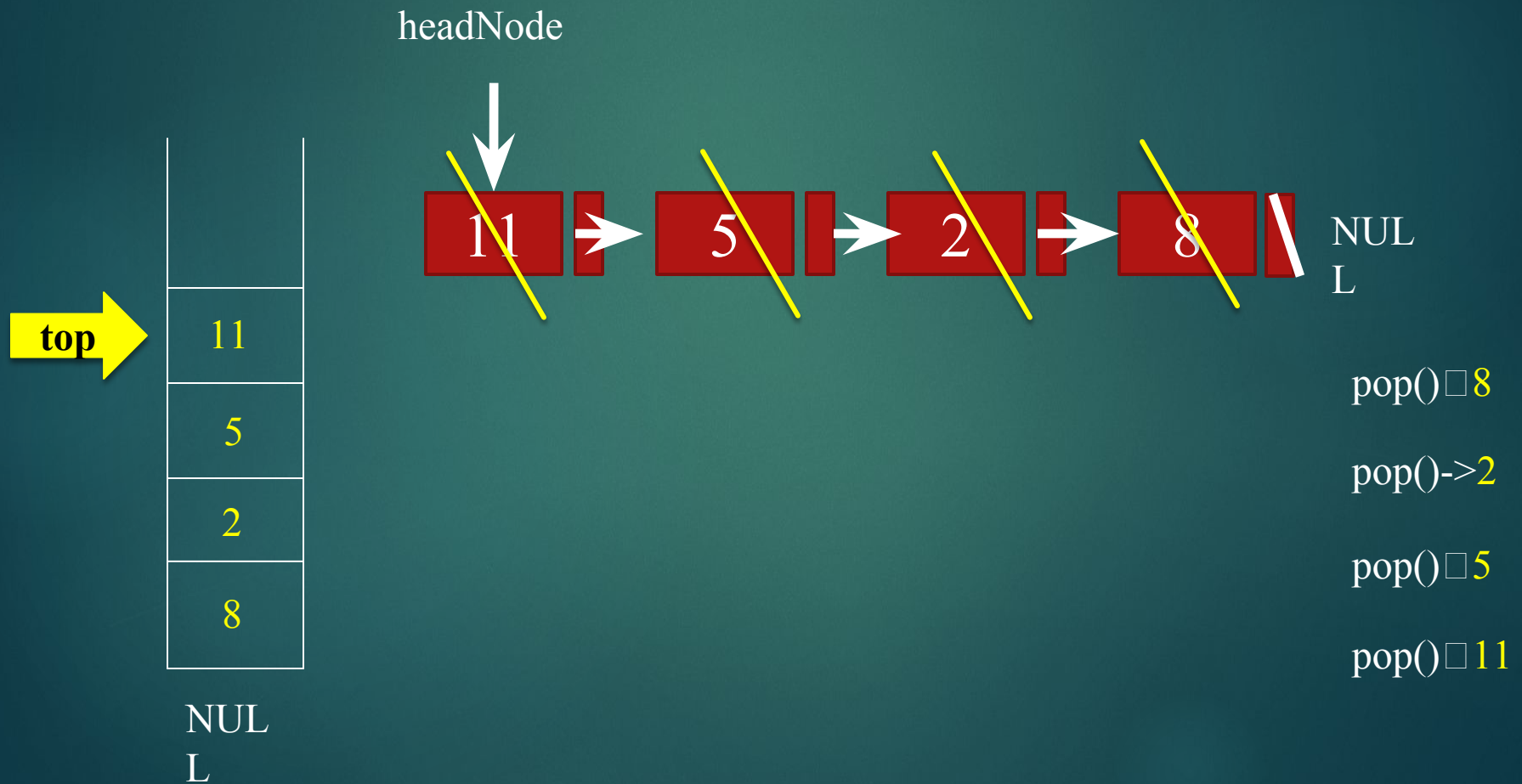


- ▶ Problems in Array
 - ▶ Size limitation in array.
 - ▶ Where to insert begin or end
 - ▶ Fixed by single link list
- ▶ Insert at start or end take constant time
- ▶ Removing an element from start is a constant time operation while removing from end is not.

Stack using link list



pop():



push():



NUL
L



headNode



push(8)

push(2)

push(5)

push(11)